When the Grass Is Greener

Three-shot search on an exponentiated Ornstein–Uhlenbeck landscape

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Abstract

A searcher evaluates a rugged landscape — the exponential of a stationary Ornstein–Uhlenbeck path — at three points chosen sequentially, and is paid the value at the final point: a terminal placement problem, distinct from approximating the path's maximum. The two-evaluation problem has a three-phase closed form (flee below the median; stay above a threshold; otherwise step a distance set by the mean-reversion scale). The third evaluation turns on a bracket of two observed locations with correlation ρ. For equal observed values b, every interior bracket point shares its conditional mean with an explicit exterior point while carrying conditional variance larger by the constant factor $(1+\rho)/(1-\rho) = e^{2\Theta}$, Θ the bracket rapidity — a matched-mean dominance that extends to every convex payoff by Gaussian convex ordering. The symmetric interior problem collapses to a parabola in $m = (\lambda_2 + \lambda)/(1 + \rho)$, namely $bm + \frac{C}{2}(1 - m^2)$ with $C = e^{2\Theta}$, whose clipped optimizer b/C yields the full phase diagram: the interior beats every alternative, observed locations included, exactly for $b_-(\rho) < b < e^{2\Theta}$ with b_- a quadratic root; the optimal interior point is the bracket middle only below $b = \sinh 2\tau$, with τ the half-bracket rapidity, beyond which a mirror pair of optima slides to the ends, arriving at $b = e^{2\Theta}$, where staying at an anchor takes over. Above that ceiling the grass is not greener: mean reversion caps the conditional mean of every unvisited point, and no variance bonus compensates. Numerically, the interior option is worth about one percent of expected value at most (peaking near $b = 0.83$), and most of the three-shot bracket widening is explained by the extra evaluation itself rather than by the interior option.

1 The problem: three evaluations of an exponentiated OU path

Let X be a stationary Ornstein–Uhlenbeck process on $\mathbb{R}$ with $\mathbb{E}[X_t] = 0$, unit marginal variance, and covariance kernel

$$k(s,t) \;=\; e^{-\kappa|s-t|}. \tag{1}$$

The landscape is $f(t) = \exp(X_t)$: locally continuous, mean-reverting in its logarithm, with occasional high peaks where excursions are magnified by the exponential. A searcher samples f at a first location, which we take to be $t_0 = 0$ by translation invariance, discovering $X_0 = b$. Two further evaluations at locations t_1 and then t_2 follow, each chosen with knowledge of all values so far, and the payoff is the value at the final location, $u = \mathbb{E}\,[f(t_2)]$. Conditional on the observations gathered by the time t_2 is chosen, X_{t_2} is Gaussian with some mean μ and variance ν, and

$$\log \mathbb{E}\big[e^{X_{t_2}} \,\big|\, \mathcal{F}\big] \;=\; \mu + \tfrac{1}{2}\nu, \tag{2}$$

where $\mathcal{F}$ denotes the observed locations and values. (The unconditional distribution after adaptive placement need not be Gaussian; every use of (2) below is conditional.) The searcher is thus paid for conditional mean and conditional variance in equal measure: a final location with residual uncertainty carries a lognormal bonus of half its variance.

Two conventions tidy the boundary cases. A searcher may take a location to infinity, which by mean reversion is equivalent to sampling a fresh $N(0,1)$ value independent of everything seen; and a searcher may finish at an observed location, receiving its known value. Neither changes the substance of any result.

2 Two evaluations: flee, explore, or stay

Given only $X_0 = b$, a candidate second location at distance t has $X_t \sim N\big(b\lambda,\, 1-\lambda^2\big)$ with $\lambda = e^{-\kappa t} \in (0,1]$. From (2),

$$\log \mathbb{E}\big[e^{X_t} \,\big|\, X_0 = b\big] \;=\; b\lambda + \tfrac{1}{2}\big(1-\lambda^2\big) \;=\; -\tfrac{1}{2}\,(\lambda - b)^2 + \tfrac{1}{2}\,\big(b^2+1\big)\,, \tag{3}$$

maximized over $\lambda \in [0,1]$ at $\lambda^* = \min\{\max\{b,0\},1\}$.

Proposition 1 (Two-shot policy). The optimal second location and its log-value $\zeta(b)$ are

$$t^* \;=\; \begin{cases} \infty & b<0 \quad \text{(flee)}, \\ -\frac{1}{\kappa}\log b & 0 \le b \le 1 \quad \text{(explore)}, \\ 0 & b>1 \quad \text{(stay)}, \end{cases} \qquad \zeta(b) \;=\; \begin{cases} \frac{1}{2} & b<0, \\ \frac{1}{2}\big(b^2+1\big) & 0\le b\le 1, \\ b & b>1. \end{cases} \tag{4}$$

A below-median first value is abandoned; an exceptional one is kept — already at two shots, an observed location can be the optimal final location. In between, the searcher steps to the distance at which the prior pull toward the mean exactly balances the anchor's height. The value ζ serves as the exterior benchmark throughout: by the Markov property, any placement beyond all existing observations faces a fresh copy of this problem anchored at the nearest observation.

3 Bridge moments, and what composes how

Suppose values $X_0 = b$ and $X_{t_1} = d$ are known, with bracket correlation $\rho = e^{-\kappa t_1}$ and rapidity $\Theta = \operatorname{artanh}\rho$. For an interior location let $\lambda_2 = e^{-\kappa t_2}$ and $\lambda = e^{-\kappa(t_1 - t_2)}$ be the correlations to the two ends, $\lambda_2\lambda = \rho$, with rapidities $\theta_2 = \operatorname{artanh}\lambda_2$ and $\theta = \operatorname{artanh}\lambda$. Gaussian conditioning gives the bridge moments

$$\mu \;=\; \frac{\lambda_2\big(1-\lambda^2\big)\,b + \lambda\big(1-\lambda_2^2\big)\,d}{1-\rho^2}, \qquad \nu \;=\; \frac{\big(1-\lambda_2^2\big)\,\big(1-\lambda^2\big)}{1-\rho^2}. \tag{5}$$

For $b = d > 0$ the bridge sags — the mean is below b everywhere strictly inside, pulled toward the process mean — while the variance rises from zero at the ends to a maximum at the middle.

A caution on what is additive. Correlations across consecutive intervals multiply, $\rho(t+s) = \rho(t)\,\rho(s)$, so the additive coordinate for composing intervals is $\kappa t = -\log\rho$, not the rapidity.

The rapidity earns its keep through a different identity: for equal anchors $d = b$ the bridge mean is

$$\mu \;=\; b\,\frac{\lambda_2 + \lambda}{1 + \lambda_2\lambda} \;=\; b\tanh(\theta_2 + \theta)\,, \tag{6}$$

which matches the mean of the exterior point at correlation $\tanh(\theta_2 + \theta)$ to its anchor — a mean-matching device we use next, not a composition law. The bridge variance also takes a hyperbolic product form: substituting $1 - \tanh^2 = \operatorname{sech}^2$ into (5),

$$\nu \;=\; \operatorname{sech}(\theta_2 + \theta)\operatorname{sech}(\theta_2 - \theta)\,, \tag{7}$$

while an exterior point at correlation $\lambda' = \tanh\theta'$ to its anchor has variance $1 - \tanh^2\theta' = \operatorname{sech}^2\theta'$.

4 Matched-mean dominance at a constant variance ratio

Theorem 1 (Bridge dominance, equal anchors). Let $d = b$ and let $t_2 \in (0, t_1)$ be an interior location with rapidities (θ_2, θ). The exterior location at composed rapidity $\theta' = \theta_2 + \theta$ has the same conditional mean $b\tanh(\theta_2 + \theta)$, and the ratio of conditional variances is the same at every interior point:

$$\frac{\nu_{\text{inside}}}{\nu_{\text{outside}}} \;=\; \frac{\cosh(\theta_2 + \theta)}{\cosh(\theta_2 - \theta)} \;=\; \frac{1 + \tanh\theta_2\tanh\theta}{1 - \tanh\theta_2\tanh\theta} \;=\; \frac{1+\rho}{1-\rho} \;=\; e^{2\Theta} \;>\; 1. \tag{8}$$

Consequently the interior point's conditional distribution dominates its mean-matched exterior rival in the Gaussian convex order, and its expected payoff is at least as large for every convex payoff with finite expectation, strictly so for the exponential.

Proof. The mean statement is (6). The ratio of (7) to $\operatorname{sech}^2(\theta_2+\theta)$ is $\cosh(\theta_2+\theta)/\cosh(\theta_2-\theta)$, and expanding both via $\cosh(A\pm B) = \cosh A\cosh B \pm \sinh A \sinh B$ gives $(1+\tanh\theta_2\tanh\theta)/(1-\tanh\theta_2\tanh\theta)$, which is $(1+\rho)/(1-\rho)$ because $\tanh\theta_2\tanh\theta = \lambda_2\lambda = \rho$ on the bracket. Two Gaussians with equal means and ordered variances are ordered in the convex order (the wider equals the narrower plus independent noise), and (2) is the exponential case. □

The amplification is a property of the bracket, not the position: threefold at $\rho = \frac{1}{2}$, nineteenfold at $\rho = 0.9$, unbounded as $\rho \uparrow 1$. At the bracket's ends both variances vanish; their ratio still tends to $e^{2\Theta}$. Dominance over mean-matched exterior rivals — which the theorem explicitly constructs — decides nothing by itself: the exterior player may choose any correlation, and an observed anchor may simply be kept. The object that can fail to exist is different: an *unvisited* point whose conditional mean matches the best *observed* value. Mean reversion caps unvisited conditional means strictly below a high anchor, and whether the variance bonus makes up the shortfall is precisely the question the phase diagram answers. The next section makes this exact.

5 The constrained parabola and the complete symmetric solution

Parametrize the interior by $x = \lambda_2 \in (\rho, 1)$ and set

$$C \;=\; \frac{1+\rho}{1-\rho} \;=\; e^{2\Theta}, \qquad m \;=\; \frac{x + \rho/x}{1+\rho} \;\in\; \left[\frac{2\sqrt{\rho}}{1+\rho},\, 1\right), \tag{9}$$

under which mirror points $x \leftrightarrow \rho/x$ coincide, the lower endpoint of m is the bracket middle, and, using $\left(1-x^2\right)\left(1-\rho^2/x^2\right) = (1+\rho)^2\left(1-m^2\right)$, the equal-anchor interior log-utility becomes a constrained parabola:

$$L_{\text{inside}}(m) \;=\; b\,m + \frac{C}{2}\left(1-m^2\right), \qquad m^* \;=\; \operatorname{clip}\!\left(\frac{b}{C};\; \frac{2\sqrt{\rho}}{1+\rho},\, 1\right). \tag{10}$$

This is the organizing object of the paper: the entire symmetric analysis reads off the position of the unconstrained peak b/C relative to the interval.

Theorem 2 (Pitchfork). The bracket middle is the interior optimum iff $b/C \le 2\sqrt{\rho}/(1+\rho)$, i.e.

$$b \;\le\; \frac{2\sqrt{\rho}}{1-\rho} \;=\; \sinh 2\tau, \qquad \tau = \operatorname{artanh}\sqrt{\rho}. \tag{11}$$

For $\sinh 2\tau < b < e^{2\Theta}$ the peak is interior and attained by the mirror pair

$$x_{\pm} \;=\; \frac{b\,(1-\rho) \pm \sqrt{b^2\,(1-\rho)^2 - 4\rho}}{2}, \qquad x_+x_- = \rho, \tag{12}$$

with optimal value

$$U_{\text{off}} \;=\; \frac{C}{2} + \frac{b^2}{2C} \;=\; \tfrac{1}{2}\left(b^2 e^{-2\Theta} + e^{2\Theta}\right). \tag{13}$$

For $b \ge e^{2\Theta}$ the peak $b/C \ge 1$ lies at or beyond the interval's open end: L_{inside} is increasing in m, no interior maximum is attained, and the interior supremum is the end limit b.

Proof. $dL/dm = b - Cm$ vanishes at $m = b/C$; the interval endpoints translate via (9) into the stated b-thresholds ($b/C = 2\sqrt{\rho}/(1+\rho)$ is $b = 2\sqrt{\rho}/(1-\rho)$, and $b/C = 1$ is $b = (1+\rho)/(1-\rho)$; the hyperbolic forms follow from $\tanh\tau = \sqrt{\rho}$). Solving $x + \rho/x = (1+\rho)\,b/C = b\,(1-\rho)$ gives (12), and substituting $m = b/C$ into (10) gives (13). □

Theorem 3 (Symmetric phase diagram). For $d = b \ge 0$ the best interior point strictly beats every alternative — exterior placements and observed anchors alike — precisely for

$$b_-(\rho) \;<\; b \;<\; e^{2\Theta}, \qquad b_-(\rho) = \frac{\sqrt{\rho}\left(2 - \sqrt{2\,(1-\rho)}\right)}{1+\rho}, \tag{14}$$

the lower edge being the smaller root of $b^2\,(1+\rho) - 4b\sqrt{\rho} + 2\rho = 0$. For $b \ge e^{2\Theta}$ the optimal final location is an *observed* one: staying at an anchor beats every unvisited point, interior or exterior.

Proof. If $b \ge e^{2\Theta}$, Theorem 2 gives interior supremum b, unattained, while every exterior point loses to the anchor pointwise: for $b > 1$ and any $\lambda \in [0,1)$,

$$b - \left[b\lambda + \tfrac{1}{2}\left(1-\lambda^2\right)\right] \;=\; (1-\lambda)\left[b - \tfrac{1+\lambda}{2}\right] \;>\; 0 \tag{15}$$

(the suprema coincide at b; the dominance is strict at every unvisited point): staying at an anchor wins. If $\sinh 2\tau < b < e^{2\Theta}$ the interior optimum is U_{off}, and both comparisons are one line: for $b > 1$,

$$U_{\text{off}} - b \;=\; \tfrac{1}{2}\,e^{-2\Theta}\left(b - e^{2\Theta}\right)^2 \;>\; 0, \tag{16}$$

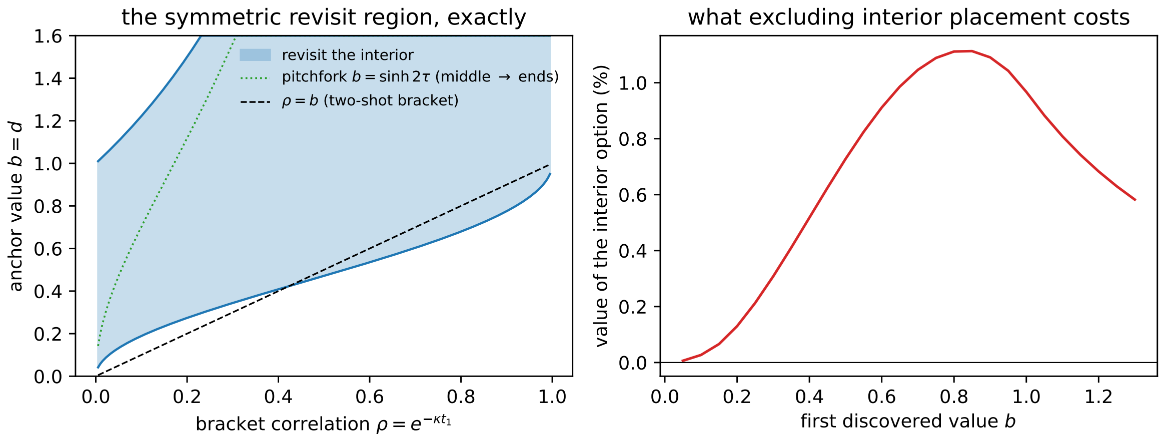

Figure 1: Left: the exact symmetric revisit region (14) in the plane of bracket correlation ρ and anchor value $b = d$, bounded below by the quadratic root b_- and above by $e^{2\Theta}$. The dotted curve is the pitchfork $b = \sinh 2\tau$: below it the optimal interior point is the bracket middle, above it the mirror pair (12) hugging the ends. The dashed line is the two-shot bracket $\rho = b$, entering the region at $b^* = 0.4233$. Right: the value of the interior option — the percentage increase in expected payoff of the full three-shot optimum over the best policy denied interior placements (it may still place on either exterior ray) — as a function of the first discovered value b, with the bracket optimized in both cases.

the arithmetic–geometric mean inequality applied to the two terms of (13), with equality only at the tangency $b = e^{2\Theta}$ defining the upper edge; for $b \le 1$,

$$U_{\text{off}} - \tfrac{1}{2}\left(b^2 + 1\right) \;=\; \tfrac{1}{2}\left(e^{2\Theta} - 1\right)\left(1 - b^2 e^{-2\Theta}\right) \;>\; 0. \tag{17}$$

If $b \le \sinh 2\tau$ the optimum is the middle, with value $U_{\text{mid}} = b\,\frac{2\sqrt{\rho}}{1+\rho} + \frac{1-\rho}{2(1+\rho)}$; for $b \le 1$, $U_{\text{mid}} > \zeta(b)$ is the stated quadratic, and $b_- \le 2\sqrt{\rho}/\left(1+\rho\right) < \sinh 2\tau$ places the lower edge inside this regime. No gap opens above the middle regime: when $\sinh 2\tau \le 1$ (i.e. $\rho \le 3 - 2\sqrt{2}$), the larger quadratic root satisfies $b_+ \ge \sinh 2\tau$, equivalent to $(1-\rho)^3 \ge 8\rho^2$, which holds on that range (left side decreasing, right increasing, valid at the endpoint); when $\sinh 2\tau > 1$, $b_+ \ge 1$ reduces, with $u = \sqrt{\rho}$, to $2u^2\left(1+u\right) \ge (1-u)^3$, true at $u = \sqrt{2} - 1$ (it reads $20\sqrt{2} \ge 28$) and increasingly so; and on $1 < b \le \sinh 2\tau$, $U_{\text{mid}} \ge b$ reduces to $b \le (1+u)\,/\,(2\,(1-u))$, which covers the stretch because $\sinh 2\tau \le (1+u)\,/\,(2\,(1-u))$ is $(1-u)^2 \ge 0$. □

Along the two-shot-inherited bracket $\rho = b$ the region opens at $b^* = 0.4233\ldots$, the root of $U_{\text{mid}}(b, b) = \zeta(b)$. The geometry of the upper edge deserves care: for $b > 1$, interior dominance requires $\rho > (b-1)\,/\,(b+1)$ — higher anchors demand a larger bracket correlation, which is a *shorter* physical bracket. Very good news is forgiven only by narrow brackets; on a wide bracket, very good news means stay.

b	optimal ρ	value	value, interior excluded	interior worth
0.1	0.053	0.8396	0.8393	0.03%
0.3	0.167	0.8723	0.8692	0.31%
0.5	0.289	0.9386	0.9314	0.73%
0.7	0.402	1.0375	1.0271	1.05%
0.83	0.469	1.1183	1.1072	1.12%
0.9	0.502	1.1670	1.1562	1.09%
1.2	0.622	1.4059	1.3991	0.68%

Table 1: The three-shot game: optimal bracket correlation, log-value, the log-value with interior placement excluded (exterior rays in both directions permitted), and the percentage the interior option adds to expected payoff. Adaptive quadrature over the second value d; interior optima exact via the stationarity quartic (18). For $b < 0$ the optimal bracket correlation tends to zero and the limiting log-value is exactly $\frac{1}{2} + \log\bigl(1 + 1/\sqrt{2\pi}\bigr) = 0.8357\ldots$

6 Beyond equal anchors

For general (b, d) the interior stationary points solve the quartic

$$x^4 - (b - \rho d)\, x^3 + \rho\, (d - \rho b)\, x - \rho^2 \;=\; 0, \tag{18}$$

obtained by differentiating (5); its real roots in $(\rho, 1)$, compared against $\zeta(b)$ and $\zeta(d)$, decide the move exactly and replace any grid. The symmetric window does not extend indefinitely in d. At $b = \frac{1}{2}$, $\rho = \frac{1}{2}$, the interior wins precisely for $d \in (\approx 0.4715,\, 1)$: below, the ray past the better end wins; at $d = 1$ every strictly interior point loses, by the identity

$$L(x) - 1 \;=\; -\frac{(2x-1)^2\,\bigl(x^2 + x + 1\bigr)}{6x^2} \;<\; 0, \qquad \tfrac{1}{2} < x < 1, \tag{19}$$

and for $d \geq 1$ staying at the second anchor is optimal — d's coefficient in the bridge mean is below one, so raising d only widens the loss. Symmetric dominance is a window, not a half-line.

7 The three-shot policy, computed

Writing $V(b, d, \rho)$ for the best of $\zeta(b)$, $\zeta(d)$ and the interior optimum (via (18)), the three-shot value of a bracket ρ is $\log \mathbb{E}_d\bigl[e^{V(b,d,\rho)}\bigr]$ with $d \sim N\bigl(b\rho, 1 - \rho^2\bigr)$, computed by adaptive quadrature and optimized over ρ. The comparison policy denies interior placement but keeps both exterior rays — it is not a ban on reversing direction.

Three observations, all in Table 1 and Figure 1. First, a below-median start still means flight, and the interior option is then worthless. Second, the three-shot bracket is wider than the two-shot step, but most of the widening is *not* bought by the interior option: at $b = 0.5$ the two-shot policy steps to $\rho = 0.5$, excluding interior placement already widens the optimum to $\rho = 0.323$, and admitting it widens further to $\rho = 0.289$. The extra evaluation and the choice it enables account for most of the stretch; the interior option adds a further, smaller pull. Third, the interior option's worth peaks near 1.1% of expected value around $b \approx 0.83$ and vanishes at both extremes. Excluding interior placement is thus cheap by value; it is the prescription that errs, and for equal anchors it errs exactly on the revisit window (14) — high, but below the ceiling $e^{2\Theta}$ above which both policies stay. For unequal observations the erring region is a joint condition on (b, d, ρ), decided by the quartic of Section 6, not membership of each value separately: at $\rho = \frac{1}{2}$, $b = \frac{1}{2}$, $d = 2$, both values lie in the symmetric window, yet staying at the second anchor is optimal and the exclusion costs nothing.

8 Discussion

The proverb, read as mathematics, is conditional, and two distinct comparisons should not be merged. Against a *visited* point of matching conditional mean, any unvisited location wins by Jensen alone: it carries variance, the known value carries none. Between two *unvisited* rivals of matching mean — an interior point and its constructed exterior counterpart — Theorem 1 gives the constant variance ratio $e^{2\Theta}$. Neither comparison manufactures a matched-mean unvisited rival for a high observed value. Mean reversion caps the conditional mean of every unvisited point, and above $b = e^{2\Theta}$ the cap binds: the optimal final location is one already seen, and the grass, there, is provably not greener. Below the cap the analysis locates the greenest patch — past the better end on bad news, in whichever direction that end lies; between the anchors on good news, at the middle for moderate anchors and hugging an end beyond the pitchfork.

A caution about the word exploration. With one evaluation remaining there is nothing to explore: no discovery at the final point can inform a later choice. The last move is pure placement, and the pull toward uncertain locations is Jensen's inequality, not curiosity. Exploration proper enters only with more shots remaining. The natural conjecture from this endgame is a two-phase k-shot policy — step out past the running best while values climb, then refine between the two best once a good region is bracketed — with the pitchfork inherited by the refinement phase, and with no irreversible switch: an adverse interior observation can make leaving attractive again. The bridge formulas make each policy evaluation cheap; the search over adaptive policies that a k-shot dynamic program requires remains to be analyzed, and we make no claim that it is small.

The problem solved here is terminal placement: the payoff is the value at the final location, not the running maximum of the path. That distinguishes it from the optimization literature on Brownian landscapes — Grill, Valko and Munos approximate the maximum of a Brownian path with many evaluations, Calvin analyzes average-case adaptive optimization, and the nearest exact few-evaluation analysis, Alabert and Caballero's laws of the minimum of a conditioned Brownian bridge, was built to compare *nonadaptive* few-point algorithms rather than to solve an adaptive placement game — and places it in the correlated-learning tradition that Callander introduced, in which each trial is a policy actually enacted and the last choice is the one lived with. Within that tradition, surveyed by Bardhi and Callander, the landscape here is the exponential of the $p = 1$ member of Bardhi's powered-exponential covariance family — Ornstein–Uhlenbeck, the member that preserves the nearest-neighbor property — and the question is a small exactly-solvable case of one their survey lists among its open methodological problems: farsighted search once restrictions on direction or horizon are dropped. Existing progress on farsighted correlated search restricts the searcher to move contiguously, choosing speed along a vein (Urgun and Yariv; Wong), or solves exact few-point placement under an information objective (Bardhi); the three-shot game here is the smallest direction-unrestricted terminal-payoff case, and its phase diagram — the constant amplification $e^{2\Theta}$, the constrained parabola (10), the pitchfork — is, to our knowledge, new.

Reproducibility

All numbers and the figure are produced by `numerics2.py` alongside this source; the review-driven identities (the constant variance ratio, the $d = 1$ factorization, the stationarity quartic, the exact flee limit) are independently checked in `verify/check_review.py`, and the earlier counterexample machinery lives in `verify/check_inside.py` and `verify/check_middle.py`, all in the browniansearch repository.

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